

```
setwd("/home/pgcp-esd/Downloads/Rdatasets/Datasets")
```

```
library(dplyr)
```

```
library(tidyr)
```

```
#####
```

```
survey <- read.csv("survey.csv", stringsAsFactor=T)
```

```
#Q1
```

```
#1
```

```
MaleNonSmokers <- survey %>%
```

```
  filter(Sex == "Male", Smoke == "Never")
```

```
#Q1
```

```
#2
```

```
PulseGT80 <- survey %>%
```

```
  filter(Pulse > 80) %>%
```

```
  select(Sex, Exer, Smoke, Pulse)
```

```
#Q1
```

```
#3
```

```
RtHand <- survey %>%
```

```
  mutate(Ratio_Hnd = Wr.Hnd / NW.Hnd) %>%
```

```
  select(Ratio_Hnd, Clap, Age)
```

```
#Q1
```

```
#4
```

```
DescStats <- survey %>%
```

```
  summarise(
```

```
    Mean_Age = mean(Age, na.rm = TRUE),
```

```
    SD_Age = sd(Age, na.rm = TRUE)
```

```
  )
```

```
#Q1
```

```
#5
```

```
DescGrp <- survey %>%
```

```
  group_by(Sex) %>%
```

```
  summarise(
```

```
    Mean_Age = mean(Age, na.rm = TRUE),
```

```
    SD_Age = sd(Age, na.rm = TRUE)
```

```
  )
```

```
#####
```

```
#Q2
```

```
items <- read.csv("items.csv", stringsAsFactor=T)
orders <- read.csv("orders.csv", stringsAsFactor=T)
ord_details <- read.csv("ord_details.csv", stringsAsFactor=T)
```

```
combined_orders <- orders %>%
  inner_join(ord_details, by = "Order.ID") %>%
  inner_join(items, by = "Item.ID")
```

```
#####
```

```
#Q3
```

```
courses <- read.csv("courses.csv", stringsAsFactor=T)
course_schedule <- read.csv("courseschedule.csv", stringsAsFactor=T)
```

```
combined_courses <- courses %>%
  inner_join(course_schedule, by = c("CourseID"="CourseCode"))
```

```
#####
```

```
#Q4
```

```
comb1 <- read.csv("comb1.csv", stringsAsFactor=T)
```

```
comb1_ <- comb1 %>%
  pivot_longer(
    cols = c(Highlighter, Marker, Pen, Refill),
    names_to = "ItemType",
    values_to = "qty"
  ) %>%
  arrange(ItemType, District)
```

```
#####
```

#Q5

```
com2 <- read.csv("comb2.csv", stringsAsFactor=T)
```

```
comb2_ <- com2 %>%  
  separate(PatientID, into = c("projectID", "temp"), sep = "-") %>%  
  separate(temp, into = c("SiteID", "PatientNumber"), sep = "/")
```

```
#####
```